Data Structures and Algorithms
Practice Questions

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Problem Statement:

Capgemini in its online written test have a coding question, wherein the students are given a string with multiple characters that are repeated consecutively. You're supposed to reduce the size of this string using mathematical logic given as in the example below :

- Input : "gabbbbeeeeffggg"
- Output: a2b4e4f2g3
- Input : "abbccccc"
- Output: ab2c5

a2b4c4f2g3

a1 ✗ ab2c5

if count > 1

s = "Adarsh"

s.charAt(2);

Ex: s = "aabbbbccccffgggg"

for ( )

s.charAt(i-1), count

count = 1 2 3 4 1 2 3 4 1 2 3

res = a2b4c4f2g3

count > 1

s.charAt(n-1), count

Ex: s = "abbccccc"

s.charAt(i-1), count

count = 1 2 3 4 5

res = ab2c5

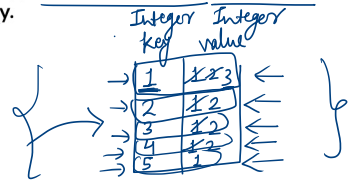
StringBuilder res = new StringBuilder();

Problem Statement:

You're given an array of integers, print the number of times each integer has occurred in the array.

Example
Input :
10 → size of the array
1 2 3 3 4 1 4 5 1 2

Output :
1 occurs 3 times
2 occurs 2 times
3 occurs 2 times
4 occurs 2 times
5 occurs 1 times



map

entry.getKey() → 1, 2, 3, 4, 5
entry.getValue() → 3, 2, 2, 2, 1

HashMap → Maintain insertion order ✗

LinkedHashMap → Maintain insertion order ✓

```
Map<Integer, Integer> map = new LinkedHashMap<>();
```

```
for (Map.Entry<Integer, Integer> entry : map.entrySet()){
    sout (entry.getKey() + " occurs " + entry.getValue() +
                                        "times");
}
```

```
for ( int num : arr) {
    map.put( num, map.getOrDefault(num, 0) + 1);
}
```

Problem Statement:

$a^3 = a * a * a$

Write a function to solve the following equation a3 + a2b + 2a2b + 2ab2 + ab2 + b3.

Write a program to accept three values in order of a, b and c and get the result of the above equation.

double a

double b

double c

$a^3 + 1a^2b + 2a^2b + 2ab^2 + ab^2 + b^3$

$= a^3 + 3a^2b + 3ab^2 + b^3$

$= a^3 + b^3 + 3ab(a+b)$

$= (a+b)^3$

$(a+b) * (a+b) * (a+b)$

or

Math.pow((a+b), 3) ⇒ Required Result

T.C. O(1)